Ecohydrology

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Eco-Hydrological Connectivity in Soil–Vegetation–Atmosphere Continuum in a Tropical Biodiversity Hotspot

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ABSTRACT

Uptake and distribution of soil water by vegetation is a dynamic phenomenon, which is of utmost importance due to its role in the hydrological cycle and terrestrial productivity. Despite the significance, the soil water–vegetation interaction remains unexplored in large parts of the world, particularly in the tropical forested regions. This study, through measurements of oxygen isotopic compositions ($\delta^{18}O$) of water in different hydrological components, such as streams, xylem, groundwater, root and soil pore water, focused on understanding the eco-hydrological connectivity during the dry season in a tropical humid forest (Western Ghats, India—a biodiversity hotspot). The results indicated that only 12% of the riparian trees primarily accessed stream water, and the dominant portion (36%) of the trees showed lower $\delta^{18}O$ than the expected (sampled) sources, indicating access to an unsampled/deeper water source, probably accessed only during the summer season to withstand low plant water potential. We explain this isotopic difference by differential access to shallow, deep and redistributed soil water, providing insights into the contribution of multiple water sources to transpiration. Hydraulic redistribution (i.e., root-mediated transport of deep water to upper soil layers) served as a critical water source for shallow-rooted plants, helping them meet their water requirements during dry periods. Evidence for limited percolation of water into deeper soil due to quick filling of shallow soil pores during short and heavy precipitation events was noticed, highlighting the importance of continuous moderate rainfall for effective water percolation into the deeper soil.

1 | Introduction

Humid tropical forests, characterized by high water content and dense vegetation, are critical in regulating the Earth's carbon and water cycle (Gentine et al. 2019; Schlesinger and Jasechko 2014). However, the predicted changes in the climate are expected to bring variations in the hydroclimatic conditions with shifts in the spatial and temporal patterns of precipitation, increased atmospheric evaporative demand (vapour pressure deficit [VPD]) and soil moisture deficits (Cook et al. 2020; IPCC 2022; Yuan et al. 2019). These shifts are likely to impact runoff and water circulation (Koirala et al. 2014), resulting in recurrent and extended drought conditions, which may reduce the transpiration rates and shift the reliance of trees towards deep-seated water/groundwater (Orlowski et al. 2023). Therefore, to predict future changes in water reliance of trees, it is prudent to understand their water sources across different climatic settings. One of the most powerful tools for tracing the water source, its movement and plant water uptake is the concurrent measurement of hydrogen and oxygen isotopic compositions (δ^2H and $\delta^{18}O$) in both xylem and source water (Dawson and Ehleringer 1991; Evaristo et al. 2015; Penna et al. 2018; Yakir and Wang 1996). The isotopic signature of source water can vary based on the extent of evaporative fractionation in the soil (Sprenger et al. 2016), percolation, subsurface mixing and recharge processes (Renée Brooks et al. 2010; Tang and Feng 2001). Root water uptake is generally a nonfractionating process (Dawson and Ehleringer 1991; Dawson and Ehleringer 1993) with the exception in case of plants that use the symplastic pathway of water movement (e.g., halophytes and xerophytes; Ellsworth and Williams 2007); hence, the source of plant xylem water can be identified by comparing with their potential sources.

In a study that sparked significant discussion, Renée Brooks et al. (2010) hypothesized the lack of hydrological connectivity between the green water (transpired water) and blue water (stream flow and groundwater)—a concept referred to as the 'Ecohydrological Separation' or 'Two Water World' hypothesis (Evaristo et al. 2015; McDonnell 2014). They proposed that plants preferentially absorb high-tension water bound within soil micropores. Later, Sprenger et al. (2016) provided an alternative explanation that this apparent separation arises from evaporative fractionation rather than distinct water pools. However, a recent finding indicates that under future warming scenarios, plants access high-tension micropore water only during the onset of rapid drying but rely on younger, more mobile water under wet conditions (Radolinski et al. 2025). While significant progress has been made in understanding plant water uptake, debates persist regarding the methodology used, particularly the influence of cryogenic extraction on ^2H fractionation (Barbeta et al. 2019; Chen et al. 2020; Newberry et al. 2017; Sprenger and Allen 2020; Zhao et al. 2016). Nonetheless, the cryogenic extraction method offers sufficient accuracy towards the utilization of $\delta^{18}O$ (H. Wang et al. 2024). To put it together, it appears important to move beyond merely determining the existence of ecohydrological separation and instead focus on understanding the processes that drive isotopic evolution within the soil–vegetation–atmosphere continuum (Sprenger and Allen 2020).

Globally, it is found that deep soil water and groundwater play a vital role in maintaining ecohydrological connectivity, especially during dry seasons (Adria et al. 2022; Barbeta and Peñuelas 2017). However, in tropical regions, ecohydrological connectivity research remains underexplored, despite its significant contribution to the global hydrological cycle (Kühnhammer et al. 2023). In India, while stable water isotope studies have provided significant insights on monsoon precipitation patterns (Lekshmy et al. 2014; Rahul, Ghosh, and Bhattacharya 2016; Saranya et al. 2021), groundwater recharge (Chidambaram et al. 2022; Pandey et al. 2023) and surface water-groundwater interaction (Saranya et al. 2020), studies on understanding ecohydrological connectivity are still in their infancy. Meteorological analyses of long-term data show that the Indian subcontinent, especially southern India, has been under extreme drought risk in the last century (Chuphal et al. 2024). Despite the drought risk, one of the southern states of India (Kerala) has been experiencing an increase in extreme rainfall events in recent decades (Shahi and Rai 2023; Sreenath et al. 2022, 2023; Vijaykumar et al. 2021). This region, bordered by the Western Ghats, one of the tropical biodiversity hotspots (Myers et al. 2000), is home to lustrous vegetation. Hence, in light of these predicted and ongoing extreme events, understanding vegetation water uptake processes and eco-hydrological connectivity in this region is crucial. Building on the existing knowledge of hydrometeorological controls on precipitation, surface water and groundwater isotopic dynamics can benefit plant water uptake studies using stable water isotopes in this region. Specifically, using $\delta^{18}O$ of water in different hydrological compartments, this study aims to understand the following: (i) the role of stream water, groundwater and the soil water towards plant water uptake in riparian species of tropical humid forest, (ii) whether elevation gradients influence $\delta^{18}O$ of plant source water and (iii) isotopic evolution of rainwater within the soil profile following precipitation in a tropical humid region.

2 | Materials and Methods

2.1 | Study Area

The Western Ghats is a north–south-oriented mountain range flanked by the Arabian Sea in the west and receives three times the average annual rainfall of India (Anika et al. 2023; Hunt et al. 2021; Krishnamurthy and Shukla 2000; Varikoden et al. 2019). This 1600-km-long mountain range with a maximum elevation of 2600 m above mean sea level (amsl) is interrupted by at least three mountain gaps (Biswas and Praveen Karanth 2021). The widest of these gaps (40 km wide), known as the Palghat Gap (average elevation of 140 m amsl), is the driest region in the state of Kerala and connects Kerala to the state of Tamil Nadu (Figure 1). Kerala, situated along the southwest corner of the Indian subcontinent, experiences rainfall brought by two monsoon systems (dual monsoon rainfall). The southwest monsoon winds, originating in the Arabian Sea, provide ~68% of the annual rainfall during June–September, while the rest is provided by the north-east monsoon and premonsoon (dry season) rainfall, bringing the long-term (1871–2005) average annual rainfall to 2817 ± 406 mm (Krishnakumar et al. 2009). During the dominant rainfall season, the southwest monsoon winds from the Arabian Sea get orographically uplifted along the Western Ghats with the eastern (leeward) side experiencing scanty rainfall (Pazhanivelan et al. 2023). The warm humid climate along with perennial rivers in the region supports enormous flora and fauna and is home to many endemic species, thus being one of the hottest biodiversity hotspots in the world (Myers et al. 2000). The vegetation diversity in the Western Ghats includes wet evergreen to semievergreen, semiclosed evergreen, stunted evergreen forests, moist deciduous, dry deciduous and savanna (Utkarsh et al. 1998). The vegetation density in the inhabited highlands of the Western Ghats includes plantation crops largely dominated by *Eucalyptus* and rubber.

2.2 | Sampling and Water Extraction

The stream water and tree core sampling for xylem water in replicates were performed along the elevation gradient of the Western Ghats during premonsoon season (April–May 2022). The pre-monsoon (driest) season was chosen for sampling to avoid interference in plant water uptake from shallow soil

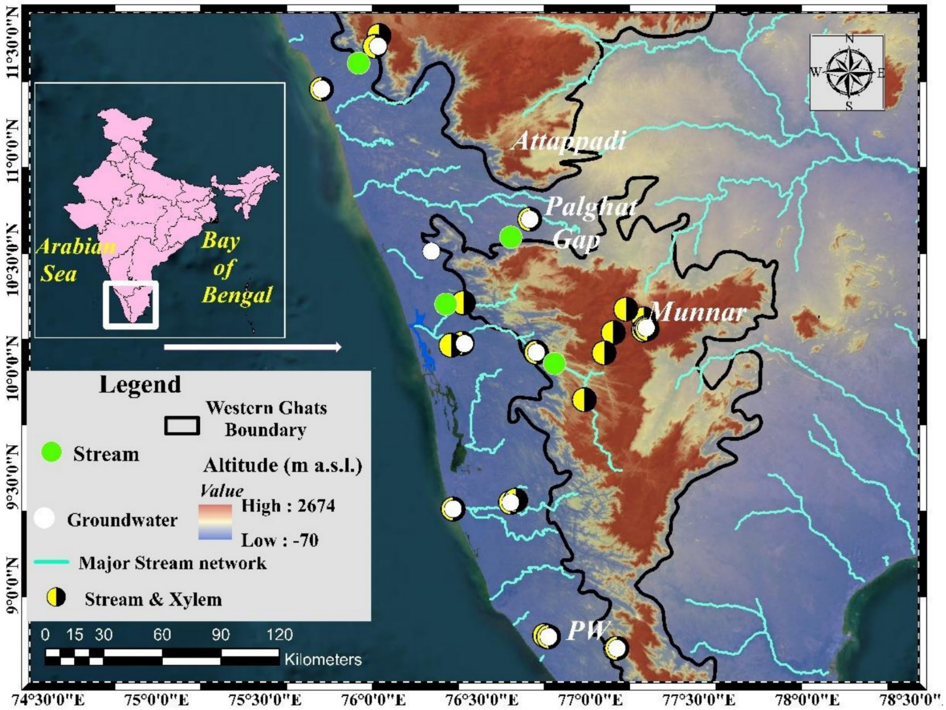

FIGURE 1 | Study area showing the sampling locations across Kerala, southern India. Soil pore water sampling was conducted at two sites: Munnar (high-elevation valley region) and PW (southernmost site). The location of Kerala within India is shown in the inset. The digital elevation model (DEM) was generated using Shuttle Radar Topography Mission (SRTM) data and overlain on Google Earth imagery using ArcMap 10.4.1.

water due to intense rainfall that could mask the signatures of deep-water uptake in trees. However, the study area received 85% excess rainfall during the sampling season owing to a few low-pressure systems. To minimize interference from recent precipitation, sampling was conducted at least 6 days after the last precipitation event. Stream sampling locations were selected based on accessibility to riparian zones and the presence of riparian trees for which sampling permission could be obtained. As the Western Ghats represent an ecologically sensitive region, tree sampling was not permitted in several areas, which restricted xylem and root (5 numbers) sampling to selected sites. Consequently, not all stream sites had corresponding tree samples. A total of 33 stream water and 33 xylem samples (Table S1) were collected (Figure 1) during the field campaign. Wherever possible, at least two healthy matured trees from the most abundant genera were chosen for sampling at each location. The maximum distance between two trees from the same sampling site was ~15 m. Wood cores were collected at the breast height of approximately 1.1 m above ground using an increment borer of 5-mm diameter and heartwood was removed by colour discrimination and by passing a light source (Fabiani et al. 2022). After removing the bark tissues, core samples were sealed in Labco exetainer vials and kept frozen until water extraction. Lateral root samples were collected from five riparian trees ($n = 5$), each located at different sampling sites, at depths of 13–15 cm below the ground level using a hand auger. Only shallow lateral roots originating in the upper soil layer were sampled to represent evaporatively fractionated topsoil water, as deeper roots may access isotopically distinct water sources. In tropical environments, kinetic fractionation in the soil is relatively minimal, and when present, is usually restricted to the top soil (Rothfuss et al. 2015; Sprenger et al. 2016). Intact root segments were gently excavated, and the adhering soil particles were quickly removed using forceps.

Twelve groundwater samples were collected from the nearby region of streams and tree samples. Apart from the data obtained during this study, $\delta^{18}O$ of precipitation and groundwater from Saranya et al. (2020) sampled along the Western Ghats of Kerala have been used for comparison. The type of wells from where groundwater was sampled included both open-dug (shallow) and deep-bore wells. The shallow-dug wells were not considered for comparison as they often experienced evaporative fractionation, while no seasonal or yearly variation was observed in deep bore wells (Figure S1). Rainwater (event-basis, $n = 4$) and soil pore water samples were also collected from one of the southern locations (PW, $n = 13$) and Munnar ($n = 1$) (Figure 1). Pore water samples were collected using suction lysimeters (Irrometer, California, USA) of varying lengths installed at 16, 24 and 37 cm below ground level (bgl). Access holes (22-mm diameter) were made to the desired depth using a soil corer to ensure a snug fit and close contact with the surrounding soil. A soil slurry was used to grout the lysimeter in place, ensuring good hydraulic contact with the soil. The soil surface around each tube was compacted to prevent preferential flow and water channelling along the tube. Soil pore water was collected daily from each lysimeter following the application of vacuum, with samples retrieved approximately at the same time. Suction lysimeters allowed the sampling of mobile soil water with little to no fractionation (Muñoz-Villers and McDonnell 2012; Geris et al. 2015). However, the pore water sample collection was successful only at certain locations and periods, as lysimeter sampling depends on the presence of sufficient mobile soil water, which was only available following precipitation events.

Precipitation accumulated over the preceding 24h was collected daily at 9:00a.m. from rooftops (~5m above ground) at the southernmost PW site using a funnel connected to a polyethylene collection vessel. Soil water sampling at the PW site was also conducted over the same duration, and the total daily precipitation was used to determine rainfall intensity.

2.3 | Laboratory Analysis

Water from xylem and root samples was extracted by heating the samples at 100°C inside a vacuum-tight tube and freezing the released water at liquid nitrogen temperature (−196°C) (Orlowski et al. 2016). The heating and extraction were carried out for an hour (West et al. 2006). The pressure in the extraction line was maintained at 0.01hPa during the extraction, which was continuously monitored using a Pirani gauge. Usually, all the xylem waters were released within 45min. This was checked by extending the heating time to 3h at 105°C with no additional release of water after 45min. The extraction yields were measured gravimetrically before and after the extraction. Efficiency, checked by oven drying the samples at 105°C for 24h, exceeded 99.7%. The moisture content of each sample was also calculated using the following:

Moisture Content = [(fresh sample—dry sample) / dry sample] × 100.

The $\delta^{18}O$ of the water samples was measured using CO_2 equilibration method (Epstein and Mayeda 1953) with a GasBench II connected to an Isotope Ratio Mass Spectrometer (MAT 253—Thermo-Fisher Scientific, Germany) at the Physical Research Laboratory, Ahmedabad, India. The precision and accuracy of the measurements were continuously monitored using a laboratory standard (Narmada River Water) with a $\delta^{18}O$ value of −4.1‰ with respect to the Vienna Standard Mean Ocean Water (Kiran Kumar et al. 2018; Rahman et al. 2021). The precision for the isotopic analysis during this study was better than 0.1‰. As such, the performance of the laboratory measurements was found to be excellent during the inter-laboratory calibration exercise conducted by the International Atomic Energy Agency (IAEA), Vienna.

2.4 | Water Enrichment at the Evaporative Site of the Leaf

Water enrichment at the evaporative site of the leaf in the sampled region has been calculated during this study. Traditionally used approximation equation (Equation 1) for estimating the isotopic enrichment at the evaporative site in the leaf (Δe) is a modified form of the Craig-Gordon model (Dongmann et al. 1974; Farquhar and Lloyd 1993).

$$\Delta e \approx \varepsilon^{+} + \varepsilon k + (\Delta \mathrm{V} - \varepsilon k)\frac{wa}{wi} \tag{1}$$

Equation (1) is an approximation of the precise form of the model (Farquhar et al. 2007)

$$\Delta e = \left(1+\varepsilon^{+}\right)\left[(1+\varepsilon k)\left(1-\frac{w_a}{w_i}\right)+\frac{w_a}{w_i}(1+\Delta V)\right]-1 \tag{2}$$

where ε^{+} is the equilibrium fractionation factor between liquid water and vapour, which was calculated following Majoube (1971); εk is the kinetic fractionation for combined diffusion through the stomata and the boundary layer (Farquhar et al. 1989). The w_a/w_i is the ratio of the water vapour mole fraction in the air relative to that in the intercellular air spaces, calculated as the saturation water vapour pressure at the air temperature (w_a) and at the leaf temperature—measured on the abaxial side of the leaf (w_i). ΔV is the isotopic enrichment of atmospheric vapour compared to source water (xylem water, δS) and is estimated as:

$$\Delta V = \frac{\delta V - \delta S}{1 - \delta S} \tag{3}$$

Here, δV is the mean $\delta^{18}O$ of ambient vapour during the premonsoon season reported for the highland station in Kerala (Lekshmy et al. 2018).

2.5 | Two-Component Mixing Model

Two-component isotopic mixing model is used to estimate the proportions of two sources ('a' and 'b') in a mixture. As the mixture is made of two sources, the sum of their fractions equals 1:

$$fa + fb = 1 \tag{4}$$

$$\delta M = fa \times \delta a + fb \times \delta b \tag{5}$$

$$fb = 1 - fa \tag{6}$$

$$fa = 1 - \frac{\delta M - \delta b}{\delta a - \delta b} \tag{7}$$

where fa and fb are the fractions of sources a and b in the mixture. δM is the isotopic composition of the mixture, and δa and δb are the isotopic compositions of a and b, respectively.

3 | Results and Discussion

3.1 | $\delta^{18}O$ Variation Among Surface, Ground, Plant and Root Water

The $\delta^{18}O$ variability observed in different hydrological pools (stream, groundwater, xylem, rain and root) showed no significant difference between streams (−4.5‰ ± 0.9‰) and xylem water (−4.5 ± 1‰) when considered across all sampling sites ($p = 0.98$, Figure 2, Table S2). Similarly, groundwater $\delta^{18}O$ did not differ significantly from those of xylem water ($p = 0.16$), except for some higher values due to evaporative enrichment from the capillary fringe zone in open dug wells due to the shallow water table (Pandey et al. 2023; Saranya et al. 2020). The shallow lateral root water of Palghat Gap showed a higher $\delta^{18}O$ value of −0.2‰, indicating evaporative enrichment due to drier weather conditions. In general, the premonsoon precipitation has shown a large variability in $\delta^{18}O$ while the surface, stream and plant water depicted rather accumulative signatures.

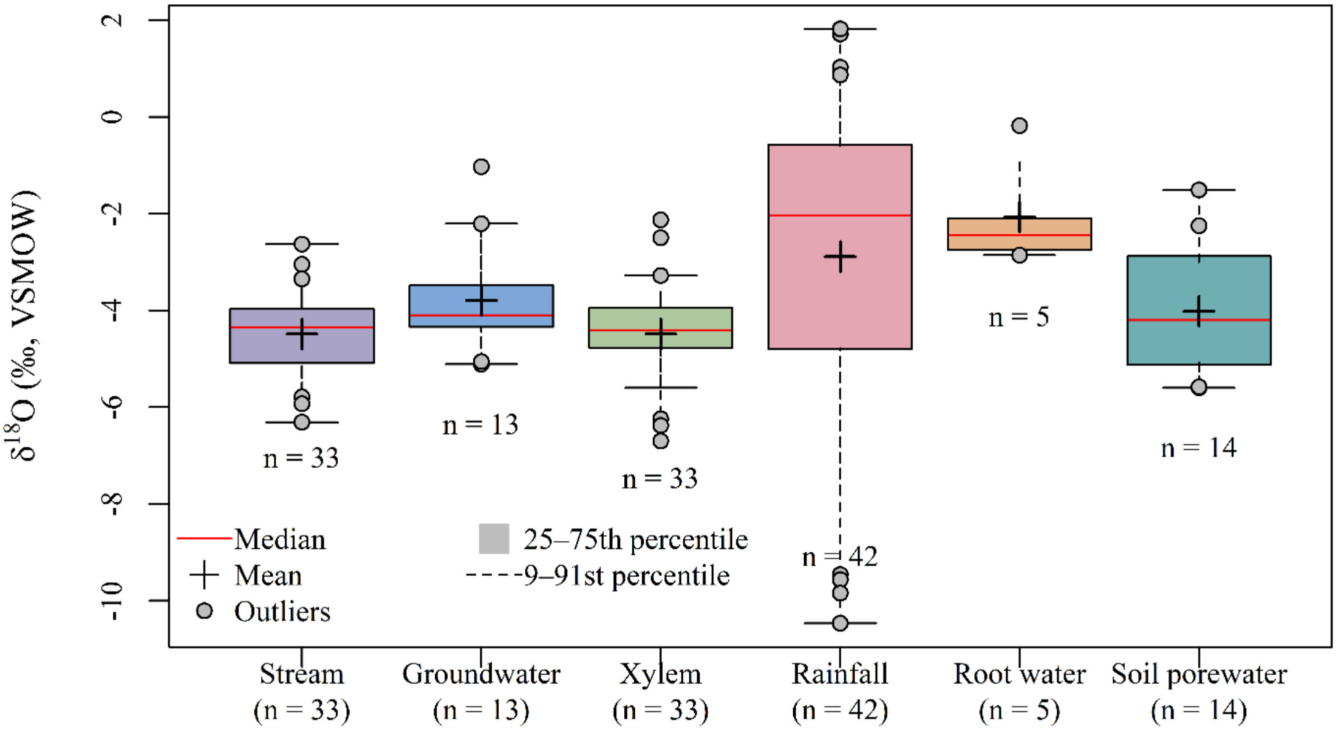

FIGURE 2 | Boxplots of $\delta^{18}O$ (‰, VSMOW) for different water pools in the study area. Boxes represent the 25th–75th percentiles with the median shown as the horizontal line; whiskers indicate the 9th and 95th percentiles (chosen to reduce the influence of extreme outliers). The mean is shown by a plus symbol, and outliers are displayed as distinct symbols. Sample size (*n*) for each water pool is indicated on the figure.

Differences in $\delta^{18}O$ of xylem water to the corresponding stream or groundwater (Δxylem) did not show any trend with elevation. The majority of the sites (19 out of 33) exhibited isotopic depletion of xylem water compared to stream water, whereas 14 sites had xylem water with higher $\delta^{18}O$ (Figure 3).

Two different trees of the same species (*Salix tetrasperma*) but of different sizes located at a higher elevation valley region (1590 m amsl; Munnar; Figure 1) exhibited contrasting xylem $\delta^{18}O$ values of—2.5 (short tree: Diameter at Breast Height (DBH) 5.7 cm and height 5.3 m) and −4.5‰ (large tree: DBH 8 cm and height 15 m), where the stream water $\delta^{18}O$ value was −5.9‰ (Figure 4). Notably, the two trees differed more in height than in DBH (common). The soil pore water and the lateral root sampled from the surface to a depth of 13 cm exhibited values of −2.3‰ and −2.4‰, respectively, which were close to the observed xylem water for the short tree (Δ xylem = −0.3‰), indicating that the short tree used water predominantly from shallow soil layers, with minimal contributions from stream water. In contrast, xylem water of the large tree (−4.5) was intermediate between stream water (−5.9‰) and shallow soil water (−2.3‰). This implies that the tree used water from different soil pools with ~40% from the shallow soil and ~60% from the stream, as estimated by two-component isotope mixing model. Possibly the water in the stream was discharged from higher elevations while the lateral seepage from the stream into the riparian soil was not sufficient to entirely replace the immobile shallow or deep soil water having distinct isotopic values (Li et al. 2023; Sprenger et al. 2019; Renée Brooks et al. 2010). This ecohydrological separation has led to the small contribution of stream water to the short tree. However, at a comparatively higher elevation point, xylem water (−5.1‰) reflected dominant moisture contribution from stream water than shallow root water (−2.8‰ and 21%).

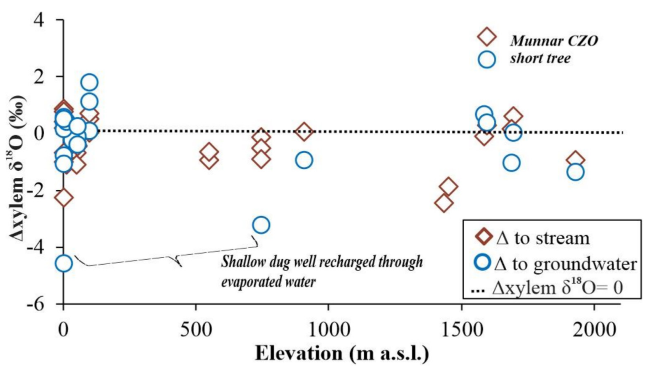

FIGURE 3 | Difference in xylem (Δxylem, ‰) water $\delta^{18}O$ to the corresponding stream (brown diamonds) and groundwater (blue circles) plotted against elevation (m a.s.l). The horizontal dotted line denotes Δxylem = 0, indicating isotopic equivalence between xylem water and the corresponding source. Positive Δxylem values indicate enrichment of xylem water relative to the source water, whereas the negative values indicate depletion.

A similar observation was made along a lower elevation site (8.76°N, 76.80°E) where two *Mangifera indica* trees of different sizes (short: 17.2 cm DBH and 7.2 m height; large: 26.4 cm DBH and 13.4 m height) exhibited different $\delta^{18}O$ values; the small one with −3.9 and the other with −4.5‰. The stream water and groundwater sampled at this site had similar $\delta^{18}O$ values (−4.08‰ and −4.1‰, respectively). The xylem water

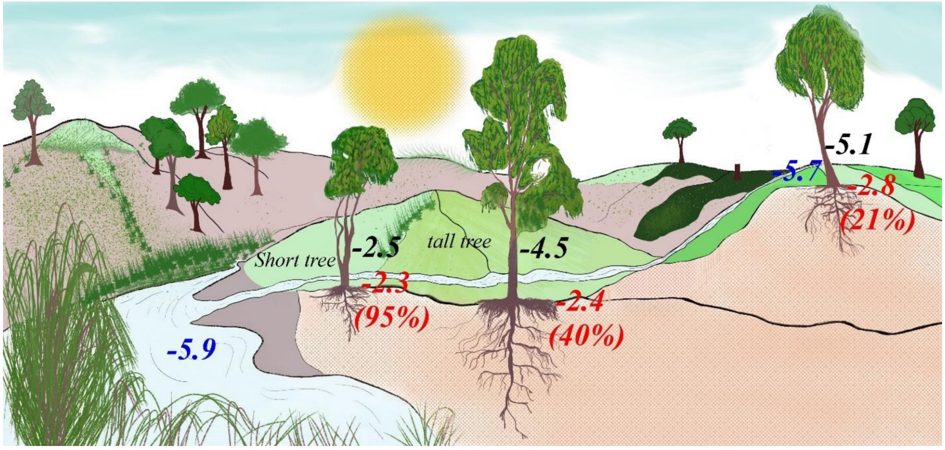

FIGURE 4 | Schematic diagram showing $\delta^{18}O$ values of water in different hydrological pools at Munnar. The blue coloured values represent stream water, the black for the xylem, and red for either soil pore water or root water samples at the topsoil (pore water and root water were not separately shown as the difference between both was negligible, 0.1‰). The numbers in the brackets represent the percentage-wise contribution of water from the topsoil to the xylem water and the remaining contribution is ascribed to the stream water.

of the short tree was slightly enriched relative to both stream water and groundwater, whereas the large tree exhibited a more depleted isotopic signature. Although we were unable to collect soil water at this site due to the absence of mobile water, the observed enrichment in the short tree is consistent with uptake from a more evaporatively enriched water source compared to stream or groundwater. The $\delta^{18}O$ value of the large tree suggests a relatively greater contribution from deeper water below the evaporation penetration limit of the soil. This pattern of tree water uptake depth, with shorter trees relying on shallow soil water, was observed in other tropical forests (Sohel et al. 2021). Short trees may not require long tap roots for anchoring in the soil; additionally, the transpiration loss from them would be less in comparison to large trees and hence may not need to access deeper water sources. Shallow soil layers are susceptible to larger fluctuations in water potential than deep soil layers or groundwater (Dawson 1996). However, this pattern might be only short-term, possibly be absent when the shallow soil moisture content gets diminished under extreme drought periods (Barbeta and Peñuelas 2017), which can either lead to drying up of the small tree or accessing deep storage water.

Two other small trees from the midland region (9.85°N and 76.97°E, elevation: 550 m amsl; Figure 1) growing on the crevices of partly weathered bedrocks located on the banks of a stream showed xylem water $\delta^{18}O$ of −4.4‰ and −4.1‰, which were depleted by −1.0‰ and −0.7‰ than the stream water (intermittent stream). The isotopic depletion of xylem water relative to stream water suggests that these trees may have accessed a water source different from the surface stream. Although rooting depth and isotopic composition of all potential water sources (deep soil water or rock moisture) were not characterized, trees growing in bedrock crevices may develop roots that penetrate fractures, enabling access to deeper rock-held water. Such water stored in weathered bedrock has been shown to act as an additional moisture source to meet the transpiration demands during dry seasons (Rempe and Dietrich 2018). Although 2022 experienced higher-than-normal rainfall, maintaining intermittent flow in this stream, it typically dries up during most summers, which substantiates the need to access deep soil/rock water.

In a rain shadow site situated ~45 km from Munnar (Figure 1), a waterfall flowing out from the protected forested regions exhibited a $\delta^{18}O$ of −5.7‰. Stream and xylem water located outside this protected region (distance of about 4 km) exhibited $\delta^{18}O$ of −4.7‰ and −4.6‰, respectively, while the shallow dug well recorded a higher value of −3.6‰. The ~1‰ and 2‰ increase in the stream water and dug well, respectively, to the waterfall supports the evaporative effect on the stream water. Conversely, groundwater from the extreme south of the study area (131 m amsl) exhibited a depleted isotopic signature (−5.1‰) compared to the nearby stream water (−4.0‰), indicating recharge from the highlands. Plant xylem water exhibited $\delta^{18}O$ of −4.0‰, −3.8‰ and 3.3‰ across different trees. Although the $\delta^{18}O$ value of shallow soil water was not available, the relatively enriched xylem water compared to the deep groundwater suggests that plants accessed a mix of water sources, including evaporatively modified near-surface water, as well as unsampled, more fractionated sources. Additionally, trees sampled from a beach area (11.29°N and 74.76°E) showed remarkably depleted $\delta^{18}O$ (−5.6‰) value, even lower than the local groundwater (−3.5‰), where the seawater $\delta^{18}O$ was 0.5‰. This indicates that the tree is likely accessing water from a recent meteoric recharge or a shallow freshwater lens rather than directly from the saturated zone. Such depleted sources are characteristic of coastal regions influenced by submarine groundwater discharge, where cool plume water (Prakash et al. 2024) commonly appears and may indicate the presence of thin low-$\delta^{18}O$ freshwater lenses above the deeper groundwater. At this location, submarine groundwater discharge has also been shown to sustain freshwater-saturated zones extending to depths of up to 20 m (Babu et al. 2021; George et al. 2018, 2021; Suresh Babu et al. 2009), suggesting that both shallow and deep freshwater with variable $\delta^{18}O$ values could be available for root water uptake in coastal environments.

3.2 | Dominant Sources of Plant Water Uptake

About 24% of the sampled trees showed hydrological connectivity with groundwater, with xylem $\delta^{18}O$ values matching groundwater. Additionally, 36% of the sampled trees displayed xylem $\delta^{18}O$ values at least 0.5‰ lower than their expected sources, with deviations reaching up to −2.5‰. This consistent depletion across species and sites suggests access to an unidentified water source (Bowling et al. 2017). Since the sampling was conducted a minimum of 6 days after the last precipitation, the only possible occurrence of such a layer is beneath the topsoil, where evaporation is limited. The potential sources could be (i) a nonmonotonic isotopic profile in the soil, (ii) an unsampled aquifer layer or rock moisture (Barbeta et al. 2019; Jasper Oshun et al. 2016; Rempe and Dietrich 2018) or (iii) fractionation during root water uptake or isotopic heterogeneity within the xylem tissue (Barbeta et al. 2019; Ellsworth and Williams 2007; Zhao et al. 2016).

The factors leading to a nonmonotonic isotopic profile in the soil profile can be diverse. Water held tightly in the soil micropores can exhibit a different isotopic composition compared to mobile water, which is accessed by the vegetation (Renée Brooks et al. 2010). In the study area, north-east (winter) monsoon precipitation (Lekshmy et al. 2015; Rahul and Ghosh 2018; Saranya et al. 2020) and the excess summer season rainfall (85% excess than usual) might have produced extremely depleted $\delta^{18}O$ in precipitation. This isotopic depletion is likely due to frequent low-pressure systems over the Bay of Bengal, ultimately allowing the storage of ^{18}O-depleted water in the subsurface layers (Good et al. 2014; Rahul, Ghosh, Bhattacharya, and Yoshimura 2016; Saranya et al. 2018). The presence of paleo-channels in the study region (Loveson et al. 2016) can lead to the incomplete mixing between fresh meteoric water and paleo-water, resulting in extremely low $\delta^{18}O$ (Jasper Oshun et al. 2016). Previously, Vargas et al. (2017) observed low $\delta^{18}O$ and $\delta^{2}H$ values in plant water and concluded it to be a result of isotopic fractionation during root water uptake similar to what is largely seen in xerophytic and halophytic plants. The preferential uptake of lighter ^{1}H by plants has also been reported (Barbeta et al. 2019; Zhao et al. 2016); however, the discrimination in oxygen isotopes is usually negligible. The current study shows a mean variation of −1.2‰ ± 0.6‰ from the sampled sources with a maximum difference of −2.3‰; which disregards the possibilities of plant-mediated fractionation. As observed elsewhere, the dominant proportion (73%, unknown source + deep water + shallow water) of the sampled trees showed isotopic difference with the respective riparian sources even during this rainfall excess pre-monsoon season (Bowling et al. 2017; Dawson and Ehleringer 1991; Wang et al. 2019). All these possibilities demand further investigation exploring the sources of this spatially widespread low xylem water $\delta^{18}O$ and if its occurrence is as a result of the extreme precipitation event or due to the sourcing from the aforementioned unknown sources.

3.3 | Elevation Effect

Generally, the isotopic elevation effect in stream water gets diminished during the dry season (pre-monsoon seasons) due to the dominance of evaporative loss and subsequent deviation of stream water $\delta^{18}O$ from the precipitation (Wu et al. 2018; Saranya et al. 2020; Jiang et al. 2021). However, despite being a dry season, a significant isotopic elevation effect in the stream water was observed in the present study, possibly due to the 85% increase in the seasonal rainfall.

The Palghat Gap region exhibited distinct $\delta^{18}O$ (higher) signatures compared to the rest of the sampling locations. The stream water isotopic elevation effect of −0.12‰/100 m along the Western Ghats reduced to −0.09‰/100 m upon removing samples from the Palghat Gap region (Figure 5). This is likely as the mountain barrier of the Western Ghats limits the chances of mixing the prevailing air masses with the leeward flows, while the Palghat Gap may be prone to mixing with dry air mass from the drier Tamil Nadu, altering the $\delta^{18}O$ signature. However, the isotopic elevation effect in the plant water is not as apparent as surface water, indicating less variation in source water for trees with elevation than surface water (present study) and groundwater (Saranya et al. 2020). Such a lack of isotopic elevation effect in xylem water could be due to the dependence of plant water on both the shallow root water that suffers evaporative fractionation (Allen et al. 2022) and the deeper soil layer that might have recorded a dampened and lagged precipitation isotopic signal (Sprenger et al. 2016; Orlowski et al. 2016; Allen et al. 2022).

In the coastal region, mean xylem $\delta^{18}O$ averaged around −4.3‰ ± 0.6‰, whereas the long-term mean $\delta^{18}O$ of precipitation reported for coastal sites (< 100 m amsl) in Kerala ranged between −2.1‰ and −2.6 ‰ (Rahul and Ghosh 2018; Saranya

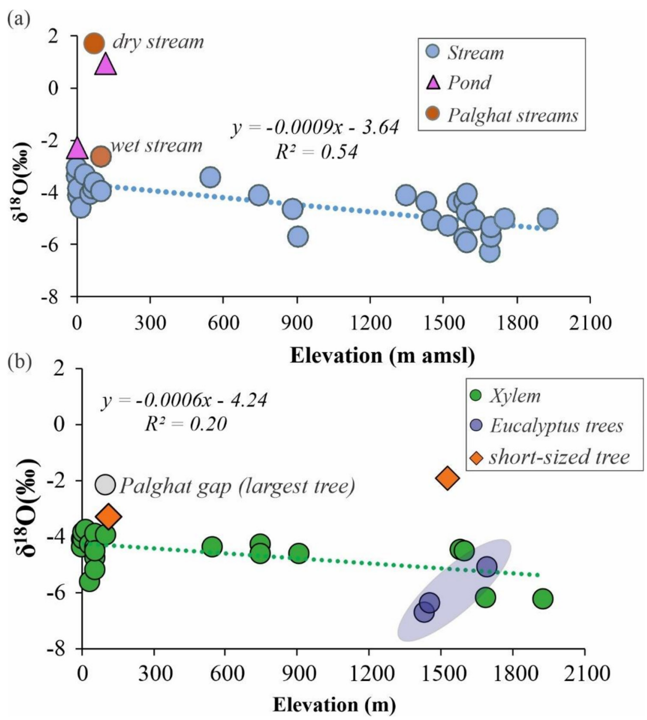

FIGURE 5 | Variation of $\delta^{18}O$ along elevation (a) in stream water and (b) plant xylem water, where the purple-filled circles are *Eucalyptus* xylem. Orange-filled diamonds are short-sized trees showing relatively higher $\delta^{18}O$ and have been omitted while calculating the elevation effect. The Palghat Gap sample were not considered for estimating the elevation effect of streams.

et al. 2021; Unnikrishnan Warrier et al. 2010). Therefore, it appears that the groundwater or deeper soil water utilized by these relatively larger trees was not directly recharged by the local precipitation. Instead, it is reflecting the integrated signatures of the past rainfall events influenced by cyclone or depression activities or groundwater recharged at higher elevations (Dansgaard 1964; Poage and Chamberlain 2001).

Evidence of water uptake from a deep/unknown source is also evident in a few *Eucalyptus* trees, which are abundant in the highlands and midlands of Kerala. The xylem water of *Eucalyptus* exhibited relatively lower $\delta^{18}O$ values compared to the other trees (Figure 5), suggesting access to deeper water sources. A previous study from Kerala reported that *Eucalyptus* can develop roots reaching depths of up to 9 m, often accessing groundwater (Kallarackal and Somen 1998). *Eucalyptus* is known to tap groundwater, particularly during the dry season, enabling it to sustain transpiration rates throughout the season (Morris et al. 2004; Liu et al. 2017; Christina et al. 2017).

3.4 | Hydraulic Redistribution Induced Masking of Rainfall Signatures in the Pore Water

Pore water sampling for 5 days at three depths (16, 24 and 37 cm bgl) along with rainfall was conducted at PW (Figure 1 and Figure 6). Generally, the pore water in the topsoil is isotopically enriched due to evaporation while water in the deeper soil layer is depleted due to the limited evaporation penetration (Sprenger et al. 2016). In this study, the pore water isotopic signature mimicked the rainfall on the first 2 days. However, despite heavy rain with a significantly lower isotopic signature on the third day, the pore water did not show a corresponding decline in $\delta^{18}O$ value and remained almost similar to the previous day (Figure 6a). It implies that the dominant fraction of the rainwater did not percolate beyond the surface zone, thus saturating the topsoil and losing the excess water as runoff. Based on the isotopic mixing model, the pore water isotopic composition, calculated assuming mixed rainwater from the second and third day of rain as the source, yielded an isotopic value of −7.7‰. This was 2.1‰ lower than the observed pore

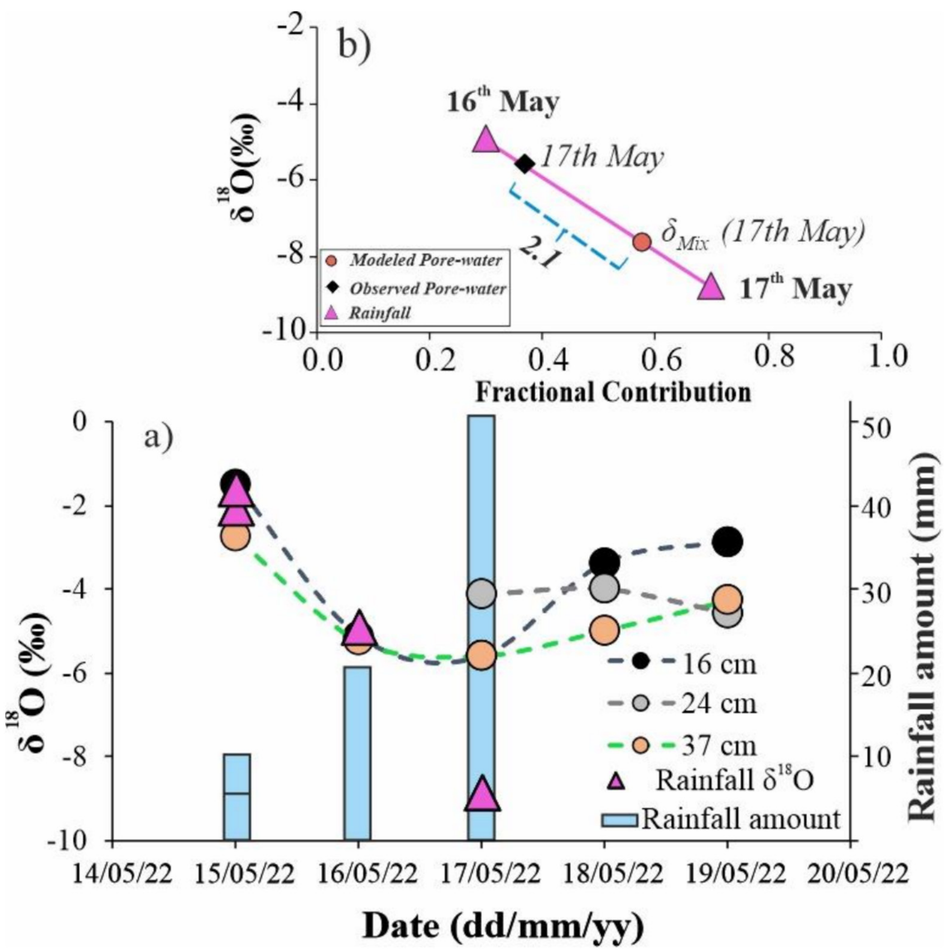

FIGURE 6 | (a) Daily variation in pore water $\delta^{18}O$ at 16 (black), 24 (grey) and 37 cm (orange) soil depths from 15th to 19th May 2022. Rainfall $\delta^{18}O$ is shown as purple triangles, and rainfall amount is shown as the light blue bars on the secondary Y-axis (right). (b) Mixing plot showing theoretical and observed $\delta^{18}O$ of pore water on 17th May in relation to rainfall $\delta^{18}O$ on the 16th and 17th May. The solid line connects successive points as a visual connector and illustrates the progression from rainwater to pore water $\delta^{18}O$.

water isotopic value on the third day, which could only be explained by the loss of precipitated water as runoff (Figure 6).

$\delta^{18}O$ values in the shallow and deeper layers were similar on the second (−5.1‰) and third days (−5.6‰), while the intermediate soil zone showed ^{18}O enrichment (−4.1‰) on the third day (Figure 7). This isotopic composition of the intermediate layer remained consistent until the fourth and fifth days, whereas the shallow and deeper layers showed an increase. This contrasting behaviour indicates that the intermediate layer was buffered by a stable water source and was disconnected from event-scale rainfall infiltration, potentially receiving root-mediated, hydraulically redistributed water (Figure 7) (Kühnhammer et al. 2023; Lee et al. 2018). This hydraulically redistributed water sustains the water potential in the drier soil layer (usually topsoil) by supplying water from the deeper layer via vertical to lateral root xylem transport.

Generally, the matrix potential tends to be the least in the top soil, which can be buffered with the hydraulic redistribution exerted by plant roots. The heavy rainfall on the third day likely generated a hydraulic gradient in the soil profile, saturating the topsoil, whereas the intermediate layer persistently received redistributed water from the deeper soil layer, probably due to the presence of large root density in this zone. Additionally, vegetation can fractionate the rainfall isotopic composition via stem flow and interception, enriching the ^{18}O in water routed to the soil (Pinos et al. 2022; Sprenger et al. 2016). When transported via preferential flow, this can create substantial isotopic differences in the pore water at certain depths. Notably, no pore water could be extracted from the intermediate soil layer on the first and second days, indicating limited mobile water and weak hydraulic connectivity at this depth. This supports the interpretation that the intermediate layer was largely isolated from precipitation infiltration and later became available by isotopically stable, redistributed water, resulting in the consistent $\delta^{18}O$ signature in the subsequent days.

3.5 | Modelling Leaf Water Isotopic Enrichment and Partitioning of Evapotranspiration

The water taken up by the plant roots gets transported to the leaves via xylem to be eventually transpired through the stomata. In this process, the leftover water on the leaf gets enriched in heavier isotopes than the source water. There is an additional isotopic exchange with the leaf and the ambient vapour (Craig and Gordon 1965) even after transpiration stops (Kim and Lee 2011; Welp et al. 2008). Cernusak et al. (2016) and Liu et al. (2023) observed a good correlation between the Craig-Gordon predicted Δe and the observed bulk leaf water enrichment. Isotopic equilibrium was also observed between the leaf water and the ambient air (in terms of relative humidity) (Cernusak et al. 2016; Cernusak et al. 2022). This shows that the calculated Δe, especially during the mid-day period when isotopic steady-state is likely to occur, may serve as a reliable estimate of the leaf water enrichment (Harwood et al. 1998).

The Δe under the steady-state condition was calculated (Table 1) at three sampling locations, that is, two highland sites (Munnar and another site 45 Km away) and the Palghat Gap (Figure 1). The lack of local meteorological variables restricted us from estimating Δe at the rest of the sites. The lowest enrichment was observed at the Palghat Gap compared to the highland sites (Table 1 & Figure 8). A strong negative relationship between Δe and relative humidity was observed, which depicted the role of atmospheric moisture in determining the transpiration. Munnar, a valley region surrounded by mountains on each side, experiences misty conditions throughout the year accompanied by higher humidity conditions (Saranya et al. 2021), thus showing lower Δe compared to the second highland site.

As leaf water isotopic enrichment is largely dependent on the ambient meteorological variables (J. Liu et al. 2023), the occurrences of saturated conditions (100% RH) recorded in the AWS data (not shown here) denote the probability of occurrences of

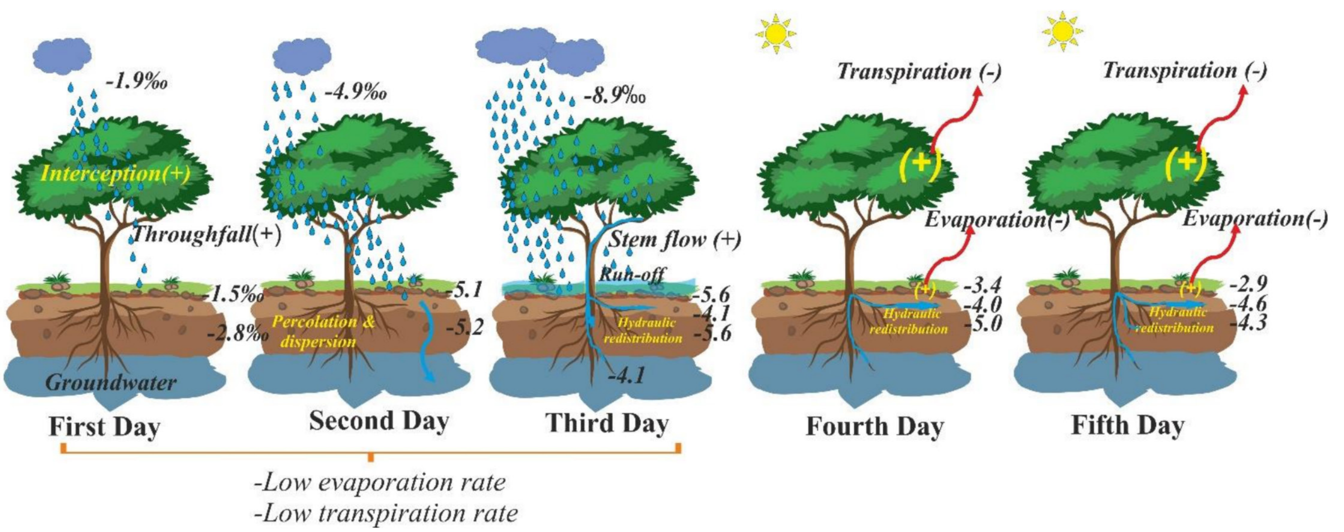

FIGURE 7 | Schematic representation of the evolution of soil pore water for five consecutive days at one of the midland sampling sites. The pore water samples were collected at depths of 16 (shallow), 24 (intermediate) and 37 (deep) cm below the ground level.

TABLE 1 | The weather conditions, source water δ^{18}O and the calculated Δe at three sites in Western Ghats.

Site	Air temperature	RH	δ^{18}O$_{Xylem}$	Predicted Δe	δ^{18}O $_{Leaf}$
Munnar	24.5	65.56	−5.09	15.19	10.10
	24.5	65.56	−4.46	14.28	9.82
	24.5	65.56	−6.24	16.18	9.94
	23.67	55.03	−2.54	17.35	14.81
	23.67	55.03	−4.51	18.40	13.89
	23.67	55.03	−6.15	19.37	13.22
45 km away from Munnar	27.34	50.71	−6.38	20.59	14.21
Palghat Gap	30	68.0	−2.13	12.99	10.86
	30	68.0	−0.21	12.08	11.87

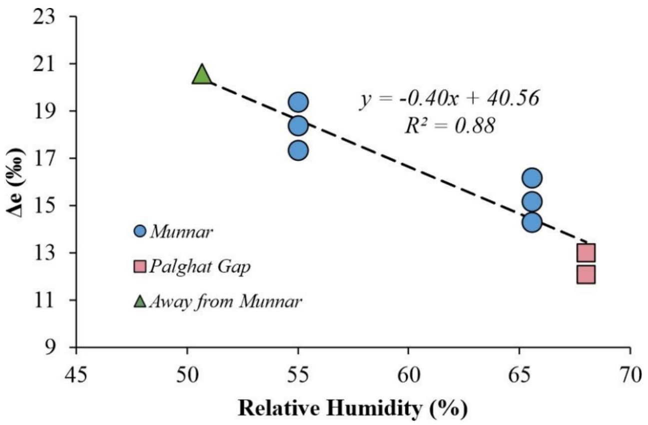

FIGURE 8 | Relationship between Craig-Gordon leaf water enrichment at the evaporative site (Δe, ‰) and Relative Humidity (%) at three study locations: Munnar (blue circles), Palghat Gap (pink squares) and another site located 45 km away from Munnar (green triangles). Δe denotes the modelled isotopic enrichment of leaf water relative to the source water under steady state conditions. Dashed lines indicate linear regression fit across the observations, with the regression equation and coefficient of determination (R^2).

isotopic equilibrium between atmospheric moisture and leaf water (Kim and Lee 2011). This is likely to happen at night when prolonged saturation conditions are present. As the atmospheric vapour isotopic composition measurements are scanty, especially in the Indian subcontinent, the measurements of bulk leaf water δ^{18}O can act as a proxy for atmospheric vapour isotopic composition. Additionally, bulk leaf water isotope data have the potential to be used as a proxy for the ambient meteorological conditions.

4 | Conclusions

This study explored hydrological connectivity in the soil–vegetation–atmosphere continuum of the tropical humid Western Ghats, a biodiversity hotspot located in southern India. The δ^{18}O in the different hydrological components has been utilized to understand the plant water source and soil water distribution processes. The results indicate differences in plant water uptake strategies across tree sizes and hydroclimatic conditions. Results showed consistently lower δ^{18}O of xylem water in large trees, while the shallow soil layers and lateral root water exhibited higher values, indicating uptake of deeper/unsampled source water by large trees. Contrary to streams, the isotopic elevation effect was not witnessed in the xylem water, suggesting waters from different soil layers to be the source for trees, thus supporting the observation of deep-water uptake. Heavy precipitation events appeared to immediately saturate the topsoil, leading to runoff and, in turn, limiting the percolation to the deep soil. Additionally, the isotopic profile of infiltrated rainfall in the soil suggested hydraulic redistribution of water from deeper to hydrologically disconnected soil layers via tree roots, maintaining moisture availability under dry spells. In the drier mountain gap region, consistently higher δ^{18}O in all the sampled water pools indicate intense evaporation effects likely extending to deeper soil layers. These findings suggest that under the projected hydroclimatic scenario, characterized by increased rainfall intensity, longer dry spells and higher atmospheric evaporative demand, shallow soil water may become increasingly transient and unavailable for plants. Consequently, plants with access to deeper water sources are likely to be resilient towards hydroclimatic variability. The observed indication of unsampled/unknown water sources further highlights the complexity of subsurface water storage in this region and underscores the need for integrated sampling of soil water, paleochannels, rock moisture and groundwater to better understand plant use strategies under varying hydroclimatic conditions. Overall findings of this study indicate that future shifts in precipitation dynamics and evaporative demand may enhance the reliance of tropical forest vegetation on deeper, more stable water sources with implications for transpiration dynamics and water cycling in the Western Ghats.

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Data Availability Statement

Data are contained within the article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** The species name, diameter at breast height (DBH) and height of the trees sampled. **Table S2:** $\delta^{18}O$ values for the different ecohydrological pools sampled in this study at each site, including stream water, xylem water and groundwater. **Figure S1:** Seasonal variation of $\delta^{18}O$ in (a) open dug well and (b) deep bore wells in the study area.